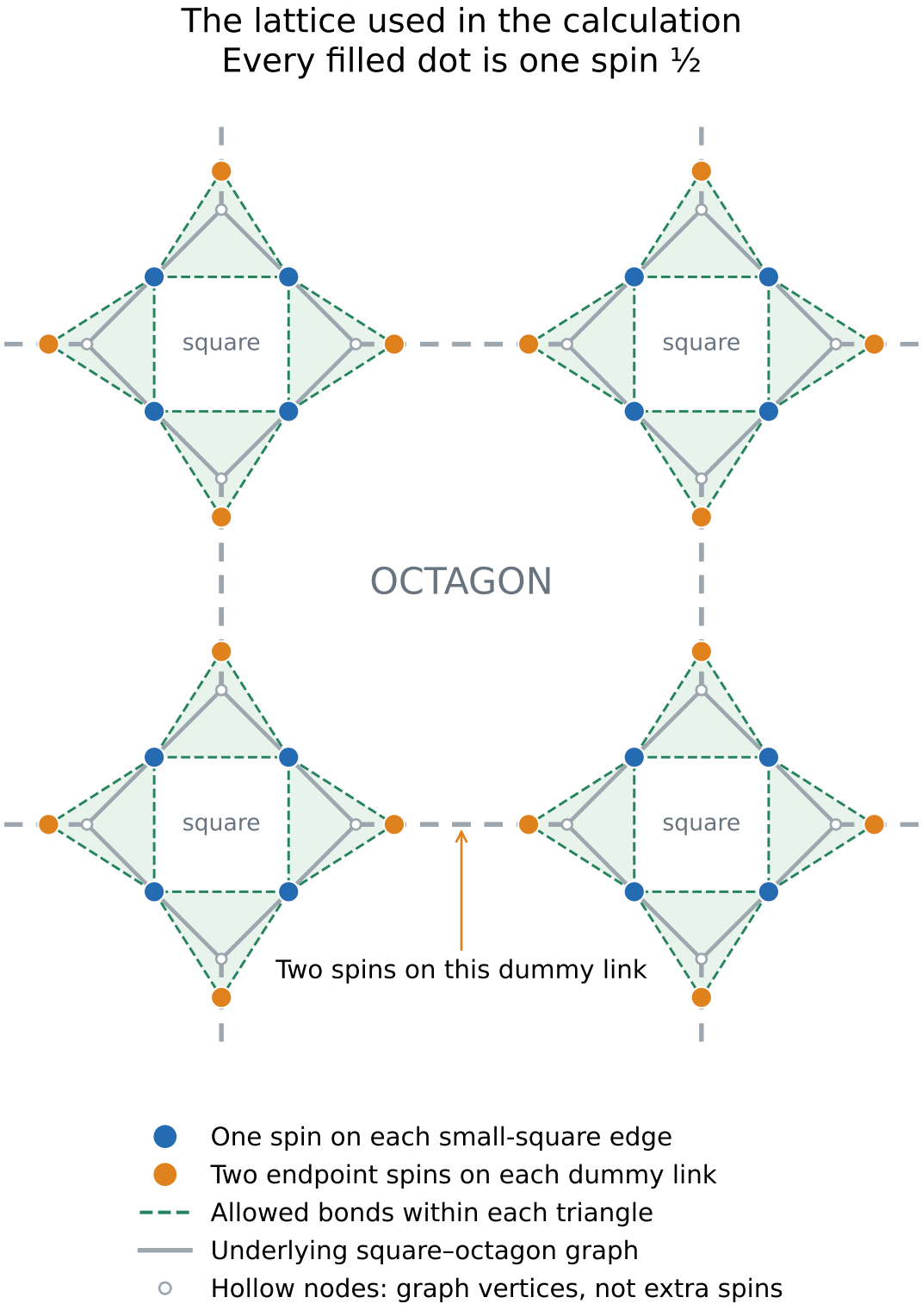
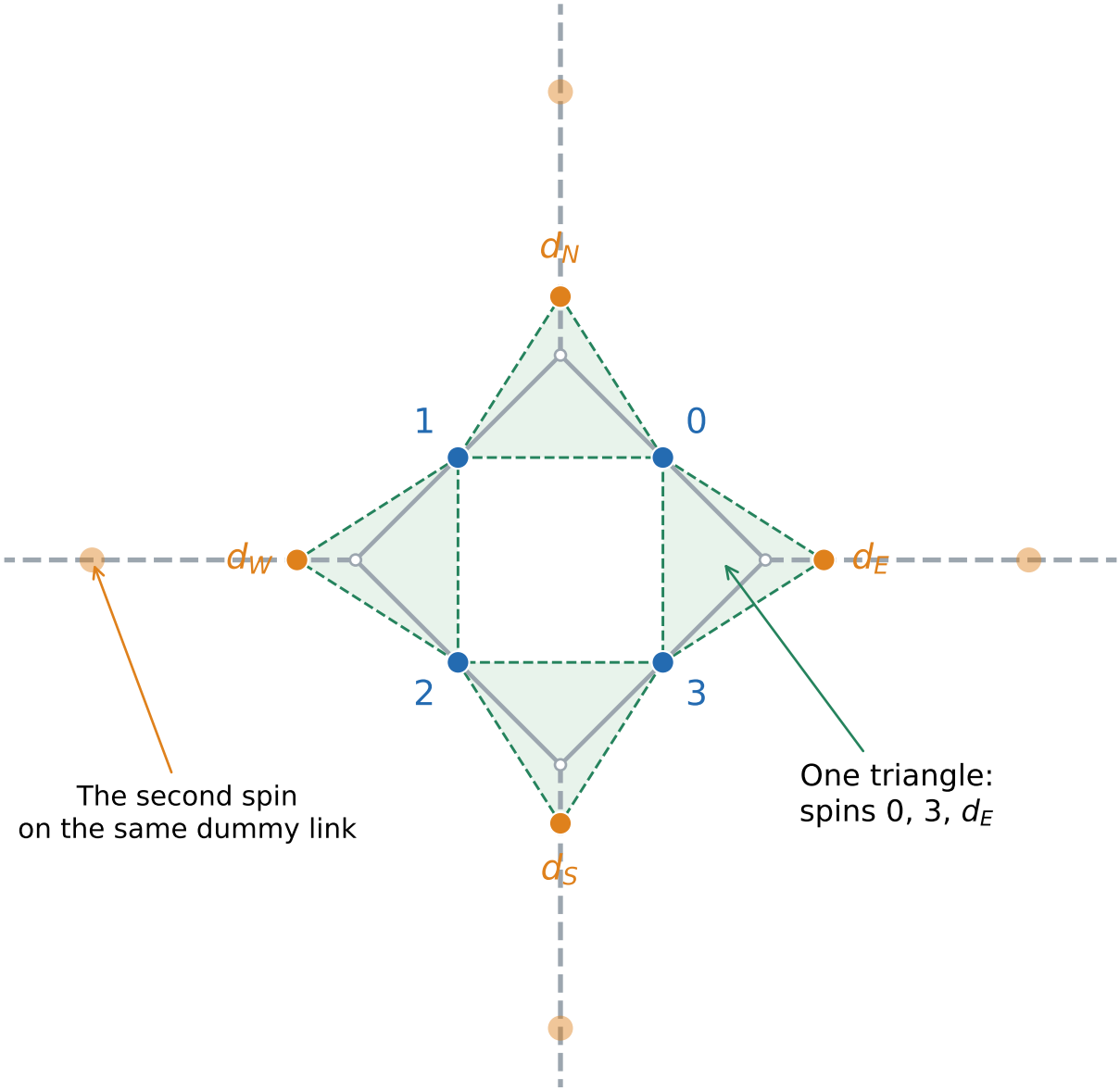
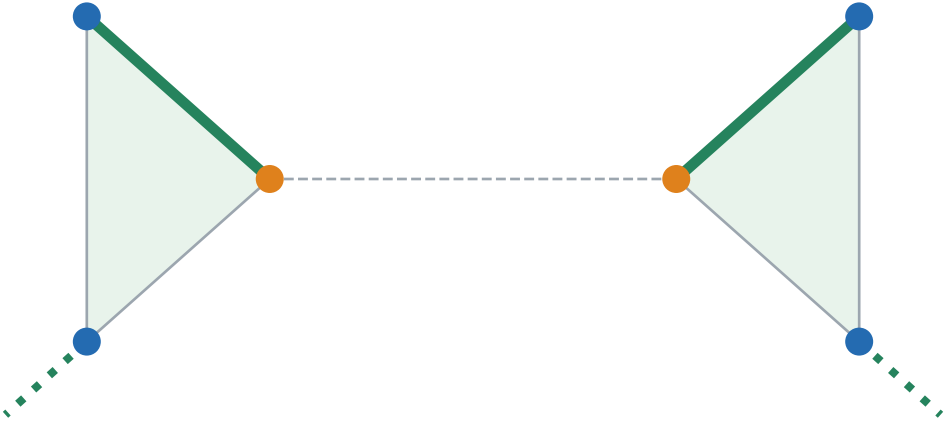

The lattice used in the calculation
Every filled dot is one spin ½
square
square
OCTAGON
square
square
Two spins on this dummy link
One spin on each small-square edge
Two endpoint spins on each dummy link
Allowed bonds within each triangle
Underlying square–octagon graph
Hollow nodes: graph vertices, not extra spins


Zoom: one simulation cell
4 blue spins + 4 orange endpoint spins = 8 spins
d_N
1
0
d_W
d_E
2
3
d_S
The second spin
on the same dummy link
One triangle:
spins 0, 3, d_E
Pale orange dots belong to neighboring cells.
Green dashed lines show allowed bonds, not a chosen covering.

Unused by triangles: the orange spins pair with each other



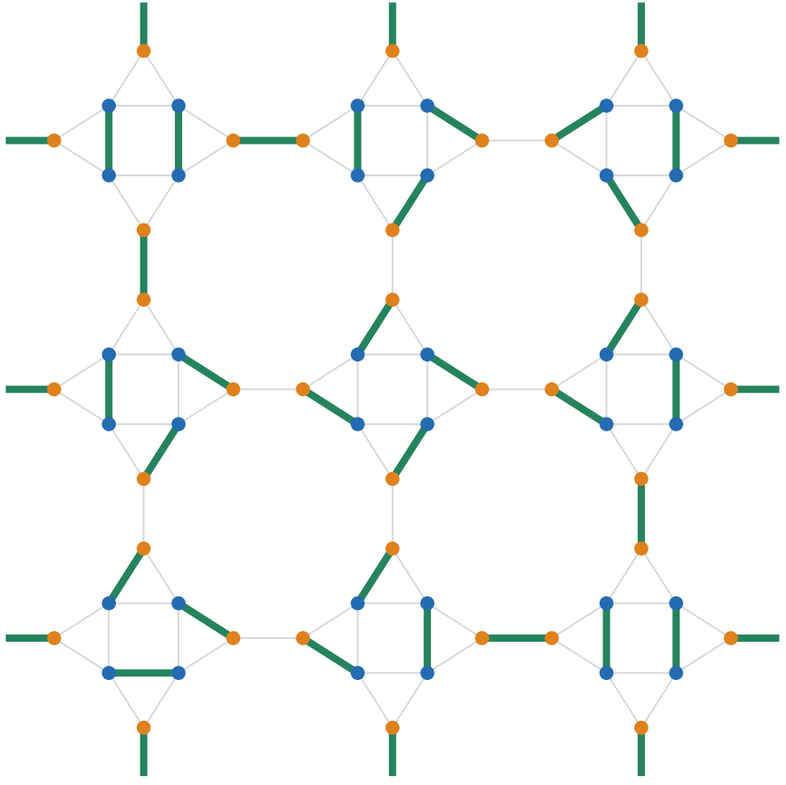
Both used by triangles: each orange spin pairs with a blue spin



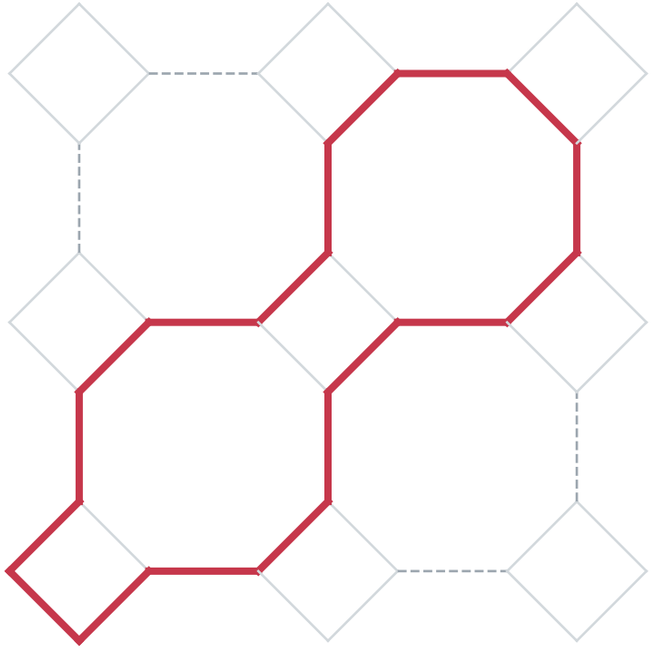
Dotted green continuations: the remaining blue spins pair in adjacent triangles.

Your mapping: covering → square–octagon loops → contract dummy links

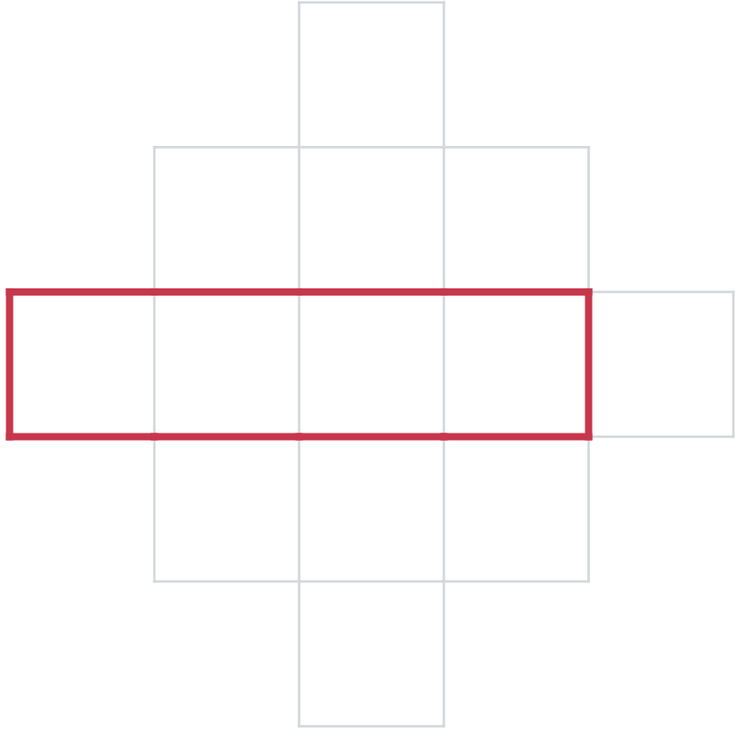
1. One allowed singlet covering



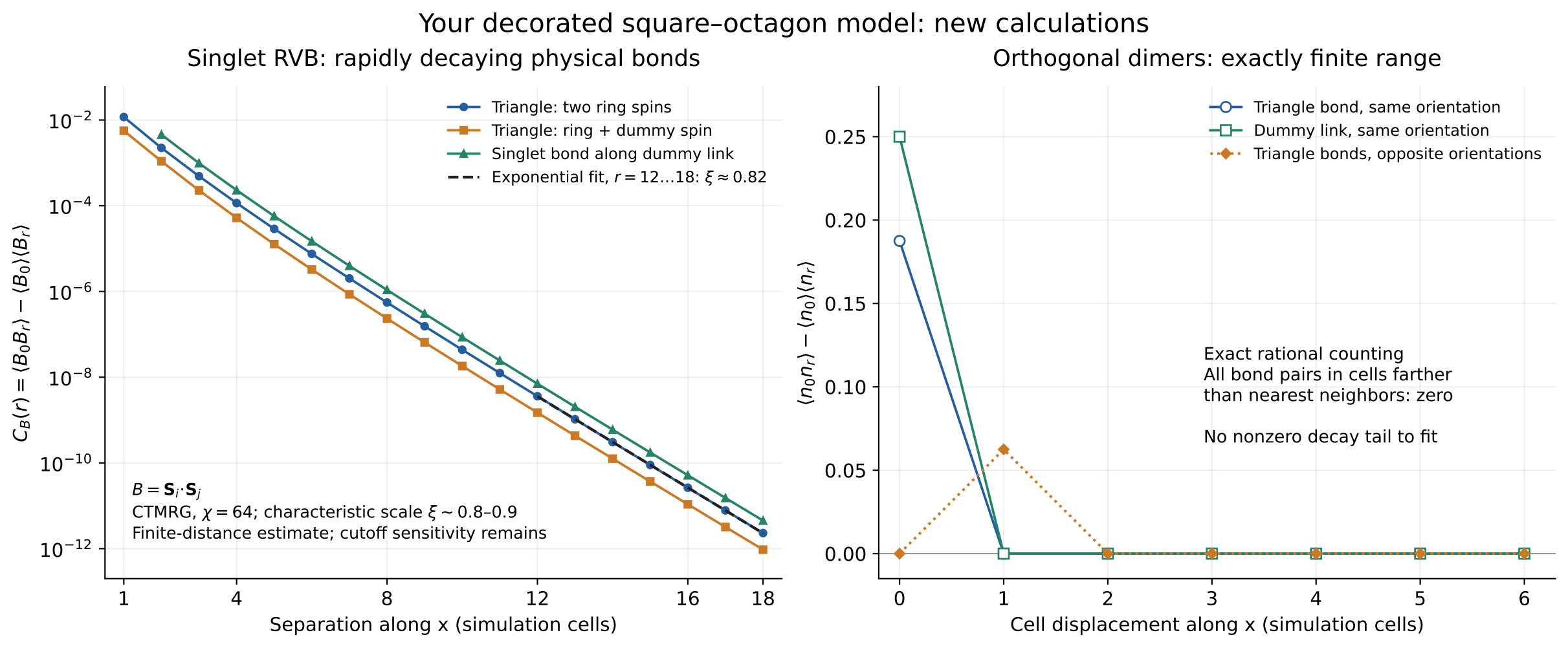
2. Loops on the square–octagon graph

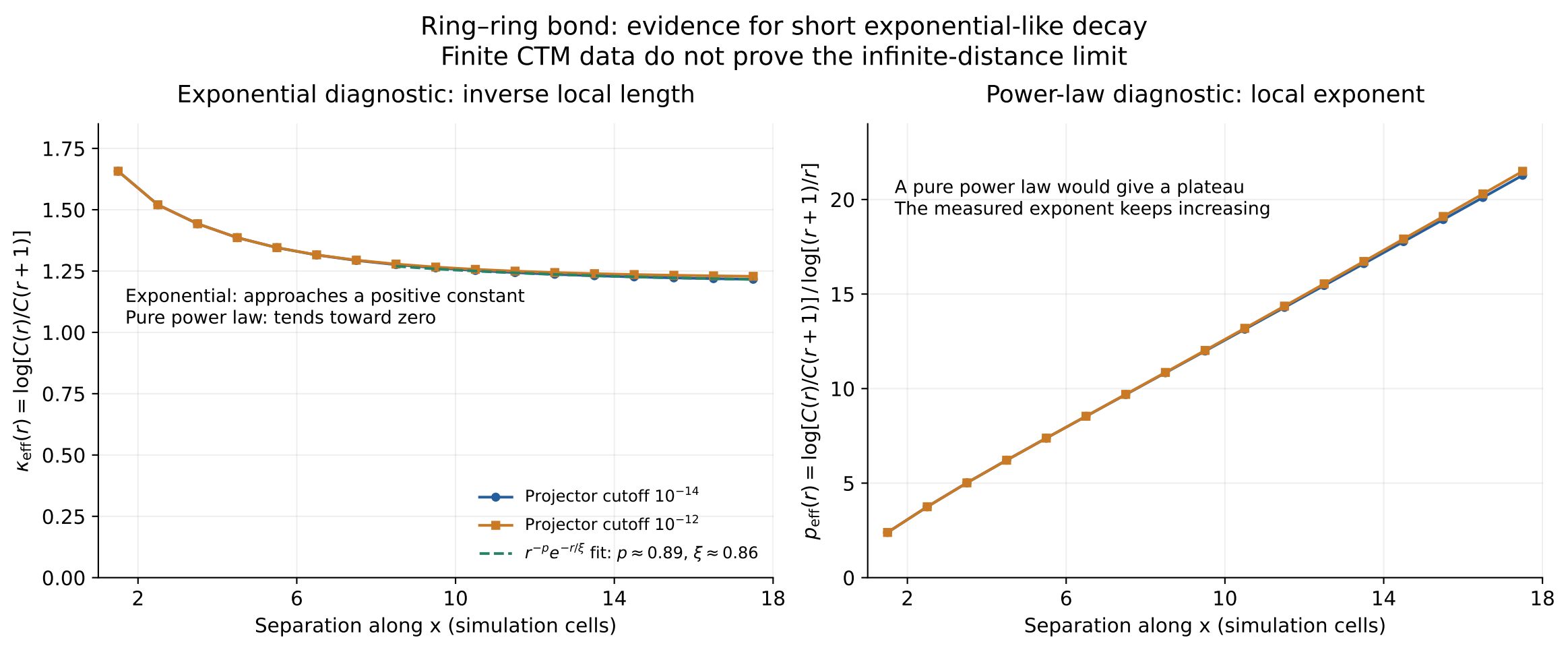


3. Shrink dummy links → square-lattice loops



The last step is one-to-one, with the inherited loop routing. It does not collapse the small squares.


Your decorated square–octagon model: new calculations
Singlet RVB: rapidly decaying physical bonds
Orthogonal dimers: exactly finite range
Triangle: two ring spins
Triangle: ring + dummy spin
Singlet bond along dummy link
Exponential fit, r = 12…18: ξ ≈ 0.82
$C_B(r) = \langle B_0 B_r \rangle - \langle B_0 \rangle \langle B_r \rangle$
$B = \mathbf{S}_i \cdot \mathbf{S}_j$
CTMRG, χ = 64; characteristic scale ξ ~ 0.8–0.9
Finite-distance estimate; cutoff sensitivity remains
Separation along x (simulation cells)
Triangle bond, same orientation
Dummy link, same orientation
Triangle bonds, opposite orientations
$\langle n_0 n_r \rangle - \langle n_0 \rangle \langle n_r \rangle$
Exact rational counting
All bond pairs in cells farther
than nearest neighbors: zero
No nonzero decay tail to fit
Cell displacement along x (simulation cells)


Ring–ring bond: evidence for short exponential-like decay
Finite CTM data do not prove the infinite-distance limit
Exponential diagnostic: inverse local length
$\kappa_{\mathrm{eff}}(r) = \log[C(r)/C(r+1)]$
Exponential: approaches a positive constant
Pure power law: tends toward zero
Projector cutoff 10^{-14}
Projector cutoff 10^{-12}
$r^{-p}e^{-r/\xi}$ fit: $p \approx 0.89$, $\xi \approx 0.86$
Separation along x (simulation cells)
Power-law diagnostic: local exponent
$p_{\mathrm{eff}}(r) = \log[C(r)/C(r+1)] / \log[(r+1)/r]$
A pure power law would give a plateau
The measured exponent keeps increasing
Separation along x (simulation cells)